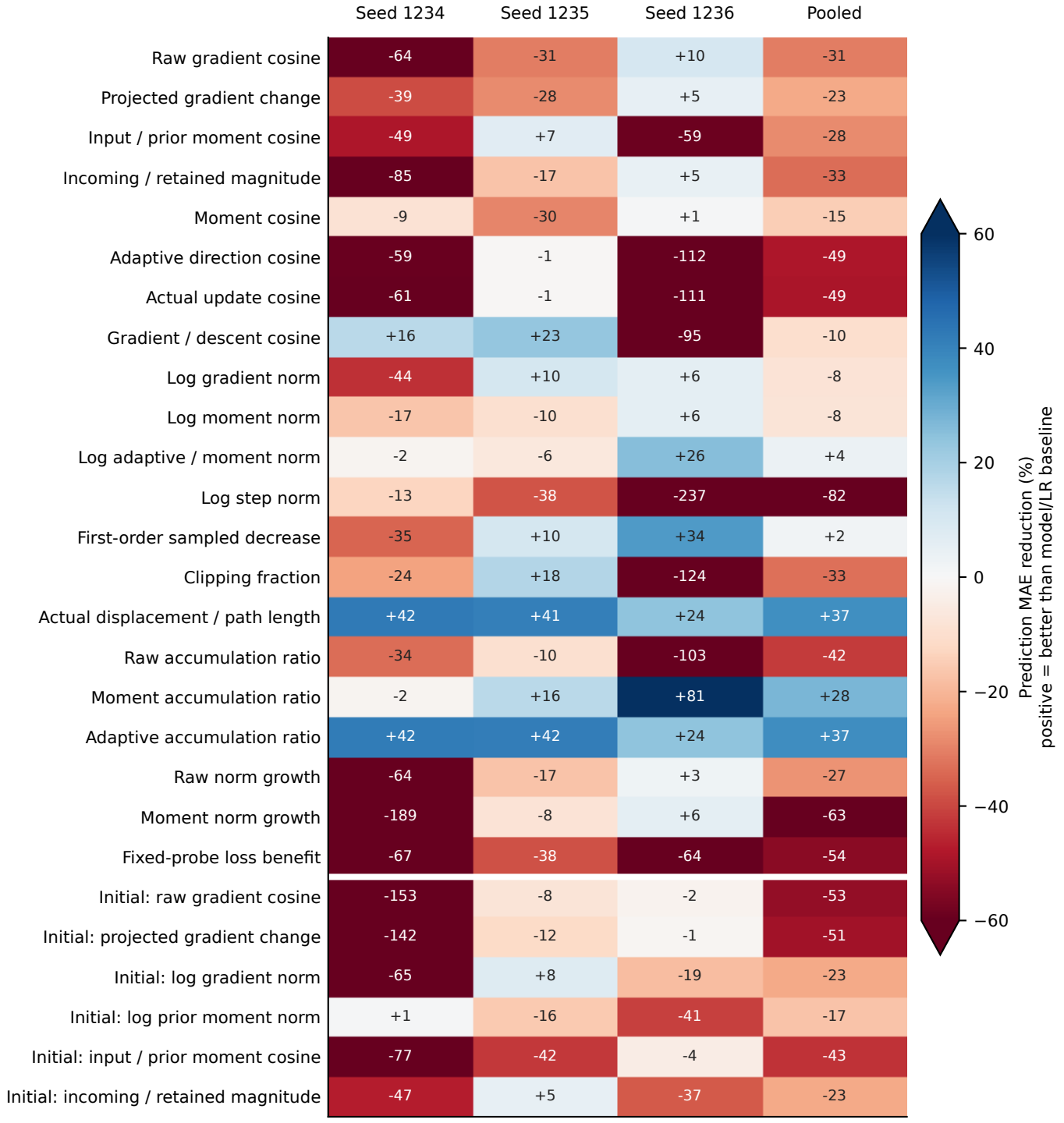

Seed 1234
Seed 1235
Seed 1236
Pooled
Raw gradient cosine
-64
-31
+10
-31
Projected gradient change
-39
-28
+5
-23
Input / prior moment cosine
-49
+7
-59
-28
Incoming / retained magnitude
-85
-17
+5
-33
Moment cosine
-9
-30
+1
-15
Adaptive direction cosine
-59
-1
-112
-49
Actual update cosine
-61
-1
-111
-49
Gradient / descent cosine
+16
+23
-95
-10
Log gradient norm
-44
+10
+6
-8
Log moment norm
-17
-10
+6
-8
Log adaptive / moment norm
-2
-6
+26
+4
Log step norm
-13
-38
-237
-82
First-order sampled decrease
-35
+10
+34
+2
Clipping fraction
-24
+18
-124
-33
Actual displacement / path length
+42
+41
+24
+37
Raw accumulation ratio
-34
-10
-103
-42
Moment accumulation ratio
-2
+16
+81
+28
Adaptive accumulation ratio
+42
+42
+24
+37
Raw norm growth
-64
-17
+3
-27
Moment norm growth
-189
-8
+6
-63
Fixed-probe loss benefit
-67
-38
-64
-54
Initial: raw gradient cosine
-153
-8
-2
-53
Initial: projected gradient change
-142
-12
-1
-51
Initial: log gradient norm
-65
+8
-19
-23
Initial: log prior moment norm
+1
-16
-41
-17
Initial: input / prior moment cosine
-77
-42
-4
-43
Initial: incoming / retained magnitude
-47
+5
-37
-23
60
40
20
0
−20
−40
−60
Prediction MAE reduction (%)
positive = better than model/LR baseline